

Business Problem statement

1) In an e-commerce platform, the objective is to analyze payment transactions to understand customer payment preferences. Using the total order amount as the basis for all calculations, the following insights are required:-

- a. Determine the total transaction value for each payment method used by customers.
- b. Calculate the average transaction amount for each payment type.
- c. Identify the maximum transaction value recorded for each payment method.
- d. Identify the minimum transaction value recorded for each payment method.
- e. Count the number of transactions carried out using each payment type.

This analysis helps in understanding the distribution of payment methods among customers and identifying patterns, trends, or anomalies in transaction behavior.

2). Categorize our customers based on the decade in which they were born. For instance, a decade '50s should encompass birth years from 1950 to 1959, while '60s should include birth years from 1960 to 1969, and so on. Ensure that the alias for each decade range is in lowercase 's'.

- Display the decade followed by the corresponding count of customers from that decade.

3). We need to provide a clear view of our top-performing customers. Our goal is to identify and showcase the top 10 customers who have spent the most with us. This information can help us recognize and appreciate our loyal customers and tailor our strategies accordingly.

4) We need to provide an overview of the payment methods we've used for transactions in the years 2020 and 2021. Specifically, we want to know:- Payment Type, Allowed, Transaction Value in 2020, Transaction Value in 2021, : This tells us the various payment methods we accept. This analysis will help us understand the performance of different payment methods over these two years and make informed decisions about their continuation.

5) To support our expansion into an offline store, we're identifying the country where we generate the highest revenue. Our sales team needs the following details of customers from that country:-

- Customer ID, First Name and Last Name, Phone Number

This information will help us establish a strong presence in the country with the highest revenue potential.

6) The objective is to analyze customer ordering behavior in the year 2021 by evaluating order frequency.

The required insights include:

- Number of orders placed (order frequency)
- Number of customers corresponding to each order frequency

This analysis helps in understanding how frequently customers place orders and reveals patterns in customer engagement and purchasing behavior.

7). The goal is to identify the top-performing customer for each payment method based on the highest order value.

The required insights include:

- Payment ID
- Customer with the highest total order amount for each payment method

This analysis helps in recognizing the most valuable customers across different payment methods and supports targeted business and marketing strategies.

8). Identify the top-selling products within each category based on their selling prices.

9) Identify which was the highest transaction value for each payment method.

This helps us gain insights into the performance of each payment method, even those that may not be active at the moment.

10.) After how many days customer placed their second order. This analysis helps us understand customer behavior and potential opportunities for retention and engagement.

11) We want to showcase the details of our customers with the maximum and minimum total spending. This analysis helps us recognize our most valuable and least engaged customers.

12). The objective is to analyze revenue trends across different cities over time by creating a structured summary.

The required insights include:

- Total revenue generated from orders
- Breakdown by city
- Organization by year and quarter based on Order Date

This analysis helps in identifying revenue patterns across locations and time periods, enabling better strategic planning and performance tracking.

13.) The goal is to identify products that have not been sold by any supplier.

This analysis helps in detecting underperforming or inactive products, allowing the business to take corrective actions such as promotions, pricing adjustments, or inventory optimization.

14) We want to understand the number of orders placed each week for each year, specifically in 2020 and 2021.

This analysis helps us identify weekly order patterns for each year, aiding in resource allocation and planning